

TASK-5

Digital Forensics – The Case of the Suspicious File

Obtain the Sample File

A suspicious file named **suspicious_download.zip** is provided for investigation. The file is collected carefully and kept unchanged because it may contain important evidence. During a forensic investigation, the original file should always be preserved to maintain its integrity.

Checksum the File

Information:

Before starting the investigation, a SHA-256 hash is generated for the file. The hash acts like a digital fingerprint and helps verify that the file has not been modified during the analysis.

Command:

```
sha256sum suspicious_download.zip
```

Explanation:

If the hash value changes later, it means the file has been altered and the evidence can no longer be considered completely reliable.

Static Analysis

Information:

Static analysis is performed without running the file. This method is safe because the file cannot affect the system while it is being examined. The

investigator looks for hidden text, URLs, or other suspicious information inside the file.

Command:

```
strings suspicious_download.exe | grep -i http
```

Explanation:

This command extracts readable text from the file and searches for web addresses or URLs that may indicate malicious activity.

Example Output:

<http://malicious-domain.com/upload.php>

Analysis:

Finding a suspicious URL like this may mean that the program is trying to communicate with an unknown server or upload data. Such behaviour is commonly seen in malware.

Dynamic Analysis**Information:**

If the file is an executable program, it can be run inside a virtual machine or sandbox. The investigator then observes its behaviour, such as creating files, changing settings, or making network connections. This helps in understanding what the program actually does

Result Analysis**Information:**

After completing the investigation, all the findings are studied carefully. The investigator determines whether the file is safe or malicious and documents the evidence collected during the analysis.

Conclusion

Digital forensics helps security professionals investigate suspicious files and understand cyber incidents. By using checksum verification and static analysis techniques, investigators can identify potential threats and collect useful evidence while keeping the original data safe.